2 **(a)** **(i)** conservation of linear momentum gives C1

$$10m + 2(5m) = mv + 4.5(5m)$$

$v = -2.5$ m s^{-1} A1

(ii) kinetic energy before collisioin $\frac{1}{2}m10^2 + \frac{1}{2}(5m)2^2 = 60m$ B1

kinetic energy after collision $\frac{1}{2}m(-2.5)^2 + \frac{1}{2}(5m)4.5^2 = 53.75m$ B1

percentage loss = $\frac{60-53.75}{60} \times 100\% = 10\%$ B1

(b) conservation of momentum gives B1

$10m + 2(5m) = mv_1 + (5m)v_2$ -------(1)

for elastic collsion, relative speed of approach = relative speed of separation. So $10 - 2 = v_2 - v_1$-------- (2) B1

Or conservation of kinetic energy gives

$\frac{1}{2}m2^2 + \frac{1}{2}(5m)2.0^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}(5m)v_2^2$ -------(2)

solving simultaneous (1) and (2) M1

$v_1 = -3.3$ m s^{-1}, $v_2 = +4.7$ m s^{-1} A1

3 **(a)** loss in gravitational potential energy = gain in kinetic energy + gain in B1